

B4 Subatomic Physics — Model Solutions, 2018

Question 1 — Reactor antineutrinos

(a) Light vs. heavy stable nuclei; reactor $\bar{\nu}$

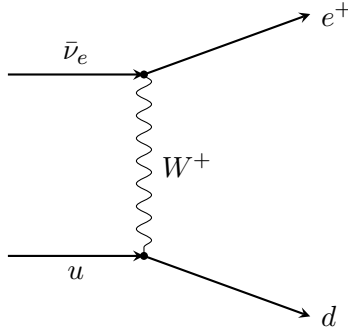
Binding from the SEMF balances surface, Coulomb, asymmetry and pairing terms. At low A , Coulomb is negligible, so the symmetric configuration $N = Z$ ($Z \approx A/2$) minimises the asymmetry term and is favoured.

For heavy nuclei the Coulomb term $\propto Z(Z-1)/A^{1/3}$ grows fast. To reduce Coulomb repulsion at fixed A , Z is shifted below $A/2$, leaving a neutron excess: stable heavy nuclei lie on the line $Z \approx A/2 - \frac{a_C}{2(a_A + a_C A^{2/3}/4)} A^{4/3}$ (qualitatively, $Z < A/2$).

Fission of a heavy nucleus produces two medium- A fragments. Inheriting the parent's neutron excess, fragments lie above the stability line for their A . They β^- -decay back ($n \rightarrow p + e^- + \bar{\nu}_e$), releasing one $\bar{\nu}_e$ per β^- -decay. Hence reactor flux is dominantly $\bar{\nu}_e$.

(b) Inverse beta decay diagram; energy dependence

Process: $\bar{\nu}_e + p \rightarrow e^+ + n$. At quark level, $\bar{\nu}_e + u \rightarrow e^+ + d$ via W^+ exchange (one of three u quarks in p):



Cross-section: in the reactor regime ($E_{\bar{\nu}} \sim \text{few MeV} \ll M_W c^2$), the W propagator is point-like, so σ is set by phase space. Energy conservation:

$$E_e = E_{\bar{\nu}} - (m_n - m_p)c^2.$$

For $E_{\bar{\nu}} \gg (m_n - m_p)c^2$ the positron is relativistic:

$$p_e c \approx E_e \approx E_{\bar{\nu}}.$$

Fermi golden-rule phase-space factor for two-body final state with one heavy spectator n :

$$\frac{d\sigma}{d\Omega} \propto p_e E_e.$$

Substitute the relativistic limit:

$$\sigma_{\bar{\nu}} \propto p_e E_e \approx E_{\bar{\nu}}^2.$$

Hence $\sigma_{\bar{\nu}}$ rises rapidly with $E_{\bar{\nu}}$.

(c) Why rate $\propto I$

Per target proton, the rate from $\bar{\nu}$ in $[E, E+dE]$ is $\Phi(E)\sigma(E)dE$, where $\Phi(E) \propto dN_{\bar{\nu}}/dE_{\bar{\nu}}$. Integrating over the spectrum:

$$R \propto \int \sigma_{\bar{\nu}}(E) \frac{dN_{\bar{\nu}}}{dE_{\bar{\nu}}} dE_{\bar{\nu}} = I.$$

(d) Numerical evaluation

(i) Fission rate. Thermal power $P = 100 \text{ MW} = 10^8 \text{ J s}^{-1}$; energy per fission $E_f = 210 \text{ MeV}$.
Convert E_f to joules using $1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$:

$$E_f = 210 \times 1.602 \times 10^{-13} \text{ J}.$$

Evaluate:

$$E_f = 3.364 \times 10^{-11} \text{ J}.$$

Fission rate $R_f = P/E_f$:

$$R_f = \frac{10^8}{3.364 \times 10^{-11}} \text{ s}^{-1}.$$

Evaluate:

$$R_f \approx 2.97 \times 10^{18} \text{ fissions s}^{-1}.$$

(ii) Integral I . Trapezoidal rule with $\Delta E = 1 \text{ MeV}$, integrand $f = \sigma_{\bar{\nu}} dN_{\bar{\nu}}/dE_{\bar{\nu}}$ taken from the table $\{0.04, 0.23, 0.30, 0.23, 0.04\}$ in $10^{-42} \text{ cm}^2 \text{ fission}^{-1} \text{ MeV}^{-1}$:

$$I = \Delta E \left[\frac{1}{2}(f_0 + f_4) + f_1 + f_2 + f_3 \right].$$

Substitute values with $\Delta E = 1 \text{ MeV}$:

$$I = 1 \cdot \left[\frac{1}{2}(0.04 + 0.04) + (0.23 + 0.30 + 0.23) \right].$$

Sum endpoints:

$$\frac{1}{2}(0.04 + 0.04) = 0.04.$$

Sum interior:

$$0.23 + 0.30 + 0.23 = 0.76.$$

Add:

$$I = 0.04 + 0.76 = 0.80 \text{ } [10^{-42} \text{ cm}^2 \text{ fission}^{-1}].$$

Convert $\text{cm}^2 \rightarrow \text{m}^2$ via $1 \text{ cm}^2 = 10^{-4} \text{ m}^2$:

$$I = 0.80 \times 10^{-42} \times 10^{-4} \text{ m}^2 \text{ fission}^{-1}.$$

$$I \approx 8.0 \times 10^{-47} \text{ m}^2 \text{ fission}^{-1}.$$

(iii) Mass of scintillator. Convert detection rate:

$$R_{\text{det}} = \frac{90}{1 \text{ hr}} = \frac{90}{3600 \text{ s}}.$$

Evaluate:

$$R_{\text{det}} = 0.025 \text{ s}^{-1}.$$

At distance d the antineutrino flux per fission is $1/(4\pi d^2)$, so:

$$R_{\text{det}} = \frac{R_f}{4\pi d^2} N_p I,$$

where N_p is the number of free protons (H atoms) in the scintillator.

Solve for N_p :

$$N_p = \frac{4\pi d^2 R_{\text{det}}}{R_f I}.$$

Geometric factor:

$$4\pi d^2 = 4\pi(7)^2 \text{ m}^2.$$

Evaluate:

$$4\pi d^2 = 4\pi(49) = 615.8 \text{ m}^2.$$

Numerator:

$$4\pi d^2 R_{\text{det}} = 615.8 \times 0.025 = 15.39 \text{ m}^2 \text{ s}^{-1}.$$

Denominator:

$$R_f I = (2.97 \times 10^{18})(8.0 \times 10^{-47}) \text{ m}^2 \text{ s}^{-1}.$$

Evaluate:

$$R_f I = 2.38 \times 10^{-28} \text{ m}^2 \text{ s}^{-1}.$$

Divide:

$$N_p = \frac{15.39}{2.38 \times 10^{-28}}.$$

Evaluate:

$$N_p = 6.47 \times 10^{28} \text{ free protons}.$$

Free protons per C_9H_{10} monomer: 10 (only H atoms; C-bound protons would be in nuclei with $Z > 1$ and so are not free targets for IBD). Molar mass of the monomer:

$$M = 9(12) + 10(1) \text{ g mol}^{-1}.$$

Evaluate:

$$M = 118 \text{ g mol}^{-1} = 0.118 \text{ kg mol}^{-1}.$$

Free protons per kg:

$$n_p = \frac{10 N_A}{M} = \frac{10 \times 6.022 \times 10^{23}}{0.118} \text{ kg}^{-1}.$$

Evaluate:

$$n_p = 5.10 \times 10^{25} \text{ kg}^{-1}.$$

Mass needed:

$$m = \frac{N_p}{n_p} = \frac{6.47 \times 10^{28}}{5.10 \times 10^{25}} \text{ kg}.$$

Evaluate:

$$m \approx 1.27 \times 10^3 \text{ kg} \approx 1.3 \text{ tonnes}.$$

Question 2 — Decuplet baryons and the Ω^-

(a) Quark-model assignments

Decuplet members are qqq with $q \in \{u, d, s\}$ in a totally symmetric flavour–spin state.

Spin. Three spin- $\frac{1}{2}$ quarks couple to $S = \frac{1}{2}$ or $\frac{3}{2}$. The decuplet picks the $S = \frac{3}{2}$ symmetric combination, giving $J = \frac{3}{2}$.

Parity. Quarks have $P = +1$, antiquarks $P = -1$. Three quarks, $L = 0$ (ground state), so $P = (+1)^3 = +1$. Hence $J^P = \frac{3}{2}^+$.

Strangeness. $S = -N_s$, where N_s is the number of s quarks: $\Delta(0s) \Rightarrow S = 0$; $\Sigma^*(1s) \Rightarrow S = -1$; $\Xi^*(2s) \Rightarrow S = -2$; $\Omega^-(3s) \Rightarrow S = -3$.

Charges. $Q_u = +\frac{2}{3}$, $Q_d = Q_s = -\frac{1}{3}$.

State	content	Q
$\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$	uuu, uud, udd, ddd	$+2, +1, 0, -1$
$\Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}$	uus, uds, dds	$+1, 0, -1$
Ξ^{*0}, Ξ^{*-}	uss, dss	$0, -1$
Ω^-	sss	-1

Symmetry concern: sss in $S = \frac{3}{2}$ ($L = 0$) is symmetric in space, spin, flavour. Pauli is rescued by the antisymmetric *colour* singlet — evidence for colour as a quantum number.

(b) Discovery reaction

$$K^-(\bar{u}s) + p(uud) \rightarrow \Omega^-(sss) + K^0(d\bar{s}) + X.$$

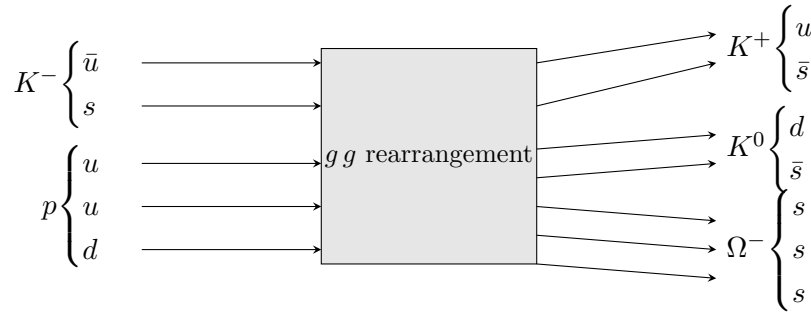
Strangeness on LHS: $S(K^-) + S(p) = -1 + 0 = -1$. RHS: $S(\Omega^-) + S(K^0) + S(X) = -3 + 1 + S(X)$.

Conservation ($\Delta S = 0$ in strong production) gives $S(X) = +1$.

Baryon number: LHS = 1, RHS without X has Ω^- ($B = 1$), so $B(X) = 0$.

Charge: LHS = $-1 + 1 = 0$. RHS without X : $-1 + 0 = -1$, so $Q(X) = +1$.

Hence X is a meson with $S = +1$, $Q = +1$: $X = K^+ (u\bar{s})$.



(Two $s\bar{s}$ pairs are created from gluons; the original s from K^- joins them to form Ω^- . The $\bar{s}$ partners pair with u and d to form K^+ and K^0 .)

(c) Mass prediction; threshold momentum

Each step along the decuplet adds one s quark. Compute the first mass step:

$$M(\Sigma^*) - M(\Delta) = 1385 - 1238 = 147 \text{ MeV}/c^2.$$

Compute the second mass step:

$$M(\Xi^*) - M(\Sigma^*) = 1532 - 1385 = 147 \text{ MeV}/c^2.$$

Linear extrapolation:

$$M(\Omega^-) \approx M(\Xi^*) + 147 \text{ MeV}/c^2.$$

Evaluate:

$$M(\Omega^-) \approx 1532 + 147 = 1679 \text{ MeV}/c^2.$$

$$\boxed{M(\Omega^-) \approx 1679 \text{ MeV}/c^2.}$$

Threshold. Lab frame: K^- on stationary p . Threshold = all final-state particles at rest in CM.
Total final invariant mass:

$$M_f = M(\Omega^-) + M(K^0) + M(K^+).$$

Use $M(K^0) \approx M(K^+) \approx 494 \text{ MeV}/c^2$:

$$M_f \approx 1679 + 494 + 494 \text{ MeV}/c^2.$$

Evaluate:

$$M_f \approx 2667 \text{ MeV}/c^2.$$

Mandelstam invariant at threshold:

$$s_{\text{th}} = M_f^2 c^4.$$

Lab expression for s :

$$s = (E_K + m_p c^2)^2 - (p_K c)^2.$$

Expand the square:

$$s = E_K^2 + 2E_K m_p c^2 + m_p^2 c^4 - (p_K c)^2.$$

Apply $E_K^2 - (p_K c)^2 = m_K^2 c^4$:

$$s = m_K^2 c^4 + m_p^2 c^4 + 2m_p c^2 E_K.$$

Equate to threshold value:

$$M_f^2 c^4 = m_K^2 c^4 + m_p^2 c^4 + 2m_p c^2 E_K.$$

Solve for E_K :

$$E_K = \frac{(M_f^2 - m_K^2 - m_p^2) c^4}{2m_p c^2}.$$

Insert $m_K = 494$, $m_p = 938$, $M_f = 2667$ (all in MeV/c^2):

$$M_f^2 - m_K^2 - m_p^2 = 2667^2 - 494^2 - 938^2 (\text{MeV}/c^2)^2.$$

Compute M_f^2 :

$$2667^2 = 7.113 \times 10^6.$$

Compute m_K^2 :

$$494^2 = 2.440 \times 10^5.$$

Compute m_p^2 :

$$938^2 = 8.798 \times 10^5.$$

Combine:

$$7.113 \times 10^6 - 2.440 \times 10^5 - 8.798 \times 10^5 = 5.989 \times 10^6.$$

Denominator:

$$2m_p c^2 = 2(938) = 1876 \text{ MeV}.$$

Divide:

$$E_K = \frac{5.989 \times 10^6}{1876} \text{ MeV}.$$

Evaluate:

$$E_K = 3192 \text{ MeV}.$$

Threshold momentum from $E_K^2 = (p_K c)^2 + m_K^2 c^4$:

$$p_K c = \sqrt{E_K^2 - m_K^2 c^4}.$$

Substitute numbers:

$$p_K c = \sqrt{3192^2 - 494^2} \text{ MeV}.$$

Compute the squares:

$$3192^2 = 1.019 \times 10^7, \quad 494^2 = 2.44 \times 10^5.$$

Subtract:

$$3192^2 - 494^2 = 9.945 \times 10^6.$$

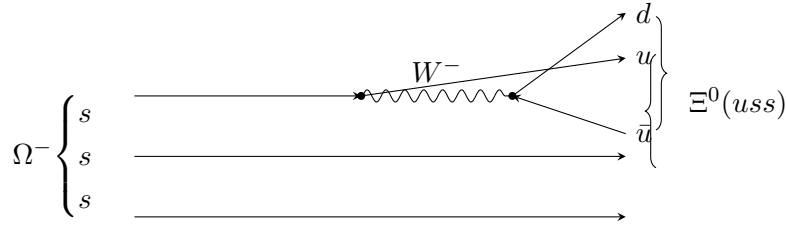
Take square root:

$$p_K c = \sqrt{9.945 \times 10^6} = 3154 \text{ MeV}.$$

$$p_{K^-}^{\min} \approx 3.15 \text{ GeV}/c.$$

(d) Decay $\Omega^- \rightarrow \Xi^0 + \pi^-$

Strangeness changes by $|\Delta S| = 1$ ($\Omega^- : S = -3 \rightarrow \Xi^0 : S = -2$): weak decay, $s \rightarrow u + W^-$, with $W^- \rightarrow d\bar{u}$.



(Spectator ss + final-state u form $\Xi^0 = uss$; $d\bar{u}$ forms π^- .)

(e) Lifetime estimate

Use measured $M_\Omega \approx 1672 \text{ MeV}/c^2$ (PDG) and $p_\Omega = 2015 \text{ MeV}/c$.

Total energy from $E^2 = (pc)^2 + (mc^2)^2$:

$$E_\Omega = \sqrt{(p_\Omega c)^2 + (M_\Omega c^2)^2}.$$

Substitute:

$$E_\Omega = \sqrt{2015^2 + 1672^2} \text{ MeV}.$$

Compute $p_\Omega^2 c^2$:

$$2015^2 = 4.060 \times 10^6.$$

Compute $M_\Omega^2 c^4$:

$$1672^2 = 2.795 \times 10^6.$$

Sum:

$$4.060 \times 10^6 + 2.795 \times 10^6 = 6.856 \times 10^6.$$

Take square root:

$$E_\Omega = \sqrt{6.856 \times 10^6} = 2618 \text{ MeV}.$$

Lorentz factor:

$$\gamma = \frac{E_\Omega}{M_\Omega c^2} = \frac{2618}{1672}.$$

Evaluate:

$$\gamma = 1.566.$$

Compute $\beta\gamma$ from momentum:

$$\beta\gamma = \frac{p_\Omega c}{M_\Omega c^2} = \frac{2015}{1672}.$$

Evaluate:

$$\beta\gamma = 1.205.$$

Solve for β :

$$\beta = \frac{\beta\gamma}{\gamma} = \frac{1.205}{1.566}.$$

Evaluate:

$$\beta = 0.770.$$

Lab speed:

$$v = \beta c = 0.770 \times 3.00 \times 10^8 \text{ m s}^{-1}.$$

Evaluate:

$$v = 2.31 \times 10^8 \text{ m s}^{-1}.$$

Lab time of flight ($L = 2.5 \text{ cm} = 0.025 \text{ m}$):

$$t_{\text{lab}} = \frac{L}{v} = \frac{0.025}{2.31 \times 10^8} \text{ s}.$$

Evaluate:

$$t_{\text{lab}} = 1.08 \times 10^{-10} \text{ s}.$$

Apply time dilation $t_{\text{lab}} = \gamma\tau$:

$$\tau = \frac{t_{\text{lab}}}{\gamma} = \frac{1.08 \times 10^{-10}}{1.566} \text{ s}.$$

Evaluate:

$$\boxed{\tau(\Omega^-) \approx 6.9 \times 10^{-11} \text{ s}.$$

(Cf. PDG $\tau \approx 8.2 \times 10^{-11} \text{ s}$ — consistent with weak decay.)

Question 3 — Baryon number, allowed reactions, GUT proton decay

(a) Baryon number assignments

- Quarks: $\mathbb{B} = +\frac{1}{3}$. Antiquarks: $\mathbb{B} = -\frac{1}{3}$.
- Leptons (and antileptons): $\mathbb{B} = 0$.
- Gauge / Higgs bosons: $\mathbb{B} = 0$.
- Baryons (qqq): $\mathbb{B} = +1$. Antibaryons ($\bar{q}\bar{q}\bar{q}$): $\mathbb{B} = -1$.
- Mesons ($q\bar{q}$): $\mathbb{B} = 0$.

(b) Allowed processes

(i) $p + p \rightarrow n + p + \pi^+$. $\mathbb{Q}$: $1 + 1 \rightarrow 0 + 1 + 1 = 2$. $\checkmark$ $\mathbb{B}$: $1 + 1 \rightarrow 1 + 1 + 0 = 2$. $\checkmark$ $\mathbb{L}$: $0 \rightarrow 0$. $\checkmark$ Strong/EM allowed (energy permitting). **Allowed.**

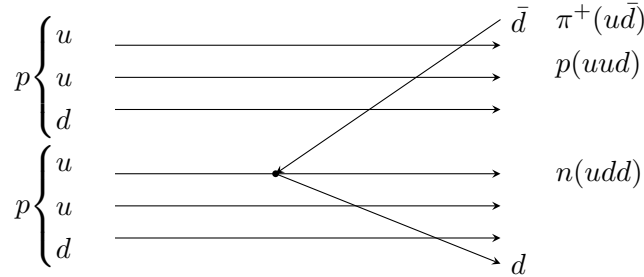
(ii) $p + e^- \rightarrow \rho^0$. $\mathbb{Q}$: $1 - 1 = 0 \rightarrow 0$. $\checkmark$ $\mathbb{B}$: $1 + 0 = 1 \rightarrow 0$. $\times$ $\mathbb{L}$: $0 + 1 = 1 \rightarrow 0$. $\times$ **Forbidden.**

(iii) $\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e$. $\mathbb{Q}$: $-1 \rightarrow -1$. $\checkmark$ $\mathbb{B}$: $0 \rightarrow 0$. $\checkmark$ L_μ : $1 \rightarrow 0 + 1 + 0 = 1$. $\checkmark$ L_e : $0 \rightarrow 1 + 0 - 1 = 0$. $\checkmark$ **Allowed** (standard muon decay).

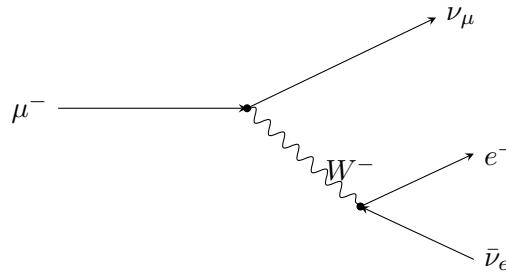
(iv) $\nu_e + e^+ \rightarrow d + \bar{u}$. $\mathbb{Q}$: $0 + 1 = 1 \rightarrow -\frac{1}{3} - \frac{2}{3} = -1$. $\times$ $\mathbb{B}$: $0 + 0 = 0 \rightarrow \frac{1}{3} - \frac{1}{3} = 0$. $\checkmark$ L_e : $1 - 1 = 0 \rightarrow 0$. $\checkmark$ **Forbidden** (charge non-conservation suffices).

Diagrams for allowed reactions:

(i) **Quark-flow for $pp \rightarrow npp\pi^+$** : Counting quarks, LHS = $4u + 2d$, RHS = $4u + 3d + \bar{d}$ so a $d\bar{d}$ pair must be created from a gluon. The new d converts one $p(uud) \rightarrow n(udd)$ by replacing a u which combines with the $\bar{d}$ to form $\pi^+(u\bar{d})$; the second proton is a spectator.



(iii) Muon decay:



(c) GUT diagrams for $p \rightarrow e^+ \pi^0$ and $n \rightarrow e^+ \pi^-$

Charge of X from the vertex $X \rightarrow u + d$ using $Q_u = +\frac{2}{3}$, $Q_d = -\frac{1}{3}$:

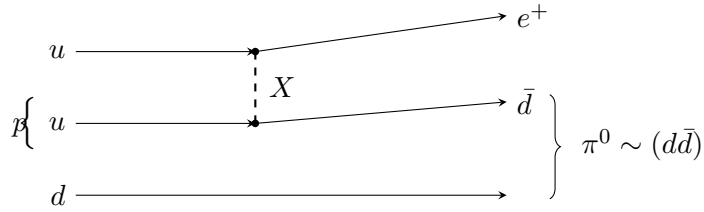
$$Q(X) = \frac{2}{3} + (-\frac{1}{3}) = +\frac{1}{3}.$$

Cross-check from $X \rightarrow e^+ + \bar{u}$ using $Q_{e^+} = +1$, $Q_{\bar{u}} = -\frac{2}{3}$:

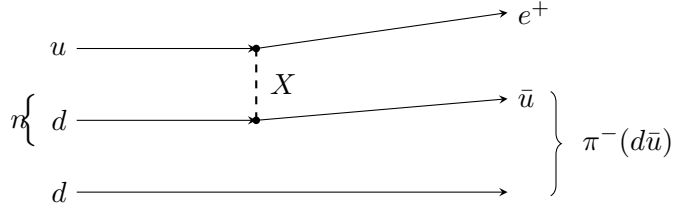
$$Q(X) = (+1) + (-\frac{2}{3}) = +\frac{1}{3}.$$

Consistent with the stated $|Q(X)| = \frac{1}{3}$.

$p(uud) \rightarrow e^+ \pi^0$: Two of the u quarks exchange an $\bar{X}$ propagator. By crossing the given vertices: $u \rightarrow e^+ + \bar{X}$ (from $X \rightarrow e^+ \bar{u}$) and $u + \bar{X} \rightarrow \bar{d}$ (from $X \rightarrow u + d$). The spectator d pairs with the produced $\bar{d}$ to form $\pi^0 \sim d\bar{d}$.



$n(udd) \rightarrow e^+ \pi^-$: Since $\pi^- = d\bar{u}$, route the u and one d through X . From $X \rightarrow u + d$ (reversed): $u + d \rightarrow X$. From $X \rightarrow e^+ \bar{u}$: $X \rightarrow e^+ + \bar{u}$. Combining: $u + d \rightarrow X \rightarrow e^+ + \bar{u}$. The spectator d pairs with the produced $\bar{u}$ to form $\pi^-(d\bar{u})$.



(d) Justification of $\tau_p \propto m_X^4 \hbar / (\lambda_q^2 \lambda_\ell^2 m_p^5 c^2)$

The decay amplitude has two vertices (λ_q and λ_ℓ) joined by an X propagator at $|q^2| \ll m_X^2 c^2$:

$$\mathcal{M} \sim \frac{\lambda_q \lambda_\ell}{q^2 - m_X^2 c^2}.$$

Take the low-momentum limit:

$$\mathcal{M} \approx -\frac{\lambda_q \lambda_\ell}{m_X^2 c^2}.$$

Square the matrix element:

$$|\mathcal{M}|^2 \propto \frac{\lambda_q^2 \lambda_\ell^2}{m_X^4 c^4}.$$

By analogy with Fermi theory of muon decay, the three-body decay rate is $\Gamma \sim G^2 E^5$ where G is the effective four-fermion coupling. Identify:

$$G_{\text{eff}} \sim \frac{\lambda_q \lambda_\ell}{m_X^2 c^2}.$$

Saturate E at the only available scale $m_p c^2$:

$$\Gamma_p \sim G_{\text{eff}}^2 (m_p c^2)^5.$$

Substitute G_{eff} :

$$\Gamma_p \sim \frac{\lambda_q^2 \lambda_\ell^2 (m_p c^2)^5}{m_X^4 c^4}.$$

Restore $\hbar$ so that $[\Gamma] = T^{-1}$ (i.e. divide by $\hbar$ to convert energy to inverse time):

$$\Gamma_p = \frac{1}{A} \frac{\lambda_q^2 \lambda_\ell^2 m_p^5 c^2}{m_X^4 \hbar}.$$

Invert to obtain the lifetime:

$$\tau_p = \Gamma_p^{-1} = A \frac{m_X^4 \hbar}{\lambda_q^2 \lambda_\ell^2 m_p^5 c^2}.$$

Power counting. The m_X^4 comes from $|1/\text{propagator}|^2$, with each $1/m_X^2$ in $\mathcal{M}$. $\lambda_q^2 \lambda_\ell^2$ from the two squared vertices. m_p^5 comes from the Fermi E^5 scaling of a three-body phase space when the only available energy scale is m_p . The factor $\hbar/c^2$ supplies the correct dimensions of time.

Question 4 — Born scattering and Higgs exchange

(a) Differential cross-section, definitions of $\mathbf{q}$, m

$d\sigma/d\Omega$ = (number of particles scattered into solid angle $d\Omega$ per unit time) divided by (incident flux $\times d\Omega$).

In the given Born formula:

- $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ is the momentum transfer (initial wavevector $\mathbf{k}$, final $\mathbf{k}'$). For elastic scattering, $|\mathbf{q}| = 2k \sin(\Theta/2)$.
- m is the (reduced) mass of the scattered particle.

(b) Yukawa cross-section; Coulomb limit

Compute the Fourier transform:

$$\tilde{V}(\mathbf{q}) = \int e^{i\mathbf{q}\cdot\mathbf{x}} V_0 \frac{e^{-\mu r}}{r} d^3x.$$

Use spherical coordinates with $\hat{\mathbf{z}} \parallel \mathbf{q}$:

$$\tilde{V}(\mathbf{q}) = V_0 \int_0^\infty dr r^2 \frac{e^{-\mu r}}{r} \int_0^\pi d\theta \sin \theta e^{iqr \cos \theta} \int_0^{2\pi} d\phi.$$

Do the ϕ -integral:

$$\int_0^{2\pi} d\phi = 2\pi.$$

Cancel the $1/r$ and substitute $u = \cos \theta$:

$$\tilde{V}(\mathbf{q}) = 2\pi V_0 \int_0^\infty dr r e^{-\mu r} \int_{-1}^1 du e^{iqr u}.$$

Evaluate the u -integral:

$$\int_{-1}^1 du e^{iqr u} = \frac{e^{iqr} - e^{-iqr}}{iqr}.$$

Apply Euler's formula:

$$\int_{-1}^1 du e^{iqr u} = \frac{2 \sin(qr)}{qr}.$$

Insert into the radial integrand:

$$\tilde{V}(\mathbf{q}) = \frac{4\pi V_0}{q} \int_0^\infty dr e^{-\mu r} \sin(qr).$$

Use the standard Laplace-type integral $\int_0^\infty e^{-\mu r} \sin(qr) dr = q/(\mu^2 + q^2)$:

$$\tilde{V}(\mathbf{q}) = \frac{4\pi V_0}{q} \cdot \frac{q}{\mu^2 + q^2}.$$

Cancel the q :

$$\tilde{V}(\mathbf{q}) = \frac{4\pi V_0}{q^2 + \mu^2}.$$

Substitute into the Born formula:

$$\frac{d\sigma}{d\Omega} = \left(\frac{m}{2\pi\hbar^2} \right)^2 |\tilde{V}(\mathbf{q})|^2.$$

Insert $\tilde{V}$:

$$\frac{d\sigma}{d\Omega} = \left(\frac{m}{2\pi\hbar^2} \right)^2 \frac{(4\pi V_0)^2}{(q^2 + \mu^2)^2}.$$

Combine the prefactors:

$$\left(\frac{m}{2\pi\hbar^2} \right)^2 (4\pi V_0)^2 = \frac{m^2 \cdot 16\pi^2 V_0^2}{4\pi^2 \hbar^4}.$$

Cancel π^2 :

$$\left(\frac{m}{2\pi\hbar^2} \right)^2 (4\pi V_0)^2 = \frac{4m^2 V_0^2}{\hbar^4}.$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{A}{(q^2 + \mu^2)^2}, \quad A = \frac{4m^2 V_0^2}{\hbar^4}.$$

Coulomb limit. Coulomb potential $V(r) = Z_1 Z_2 e^2 / (4\pi\epsilon_0 r)$ is the $\mu \rightarrow 0$ limit with $V_0 = Z_1 Z_2 e^2 / (4\pi\epsilon_0)$. Take $\mu \rightarrow 0$:

$$\frac{d\sigma}{d\Omega} = \frac{A}{q^4}.$$

Apply elastic kinematics $|\mathbf{q}| = 2k \sin(\Theta/2)$:

$$q^2 = 4k^2 \sin^2(\Theta/2).$$

Square:

$$q^4 = 16k^4 \sin^4(\Theta/2).$$

Insert into A/q^4 :

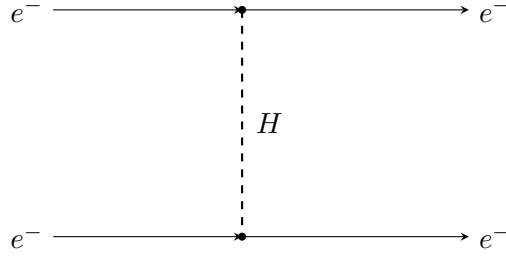
$$\frac{d\sigma}{d\Omega} = \frac{A}{16k^4 \sin^4(\Theta/2)}.$$

Rewrite using cosec:

$$\frac{d\sigma}{d\Omega} = \frac{A}{16k^4} \operatorname{cosec}^4(\Theta/2).$$

$$\boxed{\frac{d\sigma}{d\Omega} \propto \operatorname{cosec}^4(\Theta/2) \text{ (Rutherford).}}$$

(c) H vs. γ exchange



In Born/perturbation theory each single-boson exchange amplitude is (vertex)² $\times$ propagator. For a massive scalar of mass μ the Yukawa propagator gives an amplitude $\propto g^2/(q^2 + \mu^2)$; for a massless photon $\propto g_\gamma^2/q^2$.

Higgs exchange amplitude with eeH coupling λ_e :

$$\mathcal{M}_H \propto \frac{\lambda_e^2}{q^2 + \mu_H^2}.$$

Photon exchange amplitude with QED coupling $g_\gamma^2 = 4\pi\alpha$ (Heaviside–Lorentz):

$$\mathcal{M}_\gamma \propto \frac{4\pi\alpha}{q^2}.$$

Take the ratio:

$$R_H = \frac{\mathcal{M}_H}{\mathcal{M}_\gamma} = \frac{\lambda_e^2}{q^2 + \mu_H^2} \cdot \frac{q^2}{4\pi\alpha}.$$

Rearrange:

$$R_H = \frac{\lambda_e^2}{4\pi\alpha} \frac{q^2}{q^2 + \mu_H^2}.$$

Maximum value at $\sqrt{s} = 200 \text{ GeV}$. Maximum 3-momentum transfer at backscattering $\Theta = \pi$ (neglect m_e):

$$|\mathbf{q}|_{\text{max}}^2 = (2p)^2 = 4p^2.$$

For elastic massless beams in CM, $p = \sqrt{s}/2$:

$$q_{\text{max}}^2 = s = (200 \text{ GeV})^2.$$

Evaluate:

$$q_{\text{max}}^2 = 4 \times 10^4 \text{ GeV}^2.$$

Higgs mass squared:

$$\mu_H^2 = (125)^2 \text{ GeV}^2.$$

Evaluate:

$$\mu_H^2 = 1.5625 \times 10^4 \text{ GeV}^2.$$

Compute the propagator ratio:

$$\frac{q_{\text{max}}^2}{q_{\text{max}}^2 + \mu_H^2} = \frac{4 \times 10^4}{4 \times 10^4 + 1.5625 \times 10^4}.$$

Simplify:

$$\frac{q_{\max}^2}{q_{\max}^2 + \mu_H^2} = \frac{4}{5.5625} = 0.719.$$

Coupling-factor numerator:

$$\lambda_e^2 = (2.9 \times 10^{-6})^2 = 8.41 \times 10^{-12}.$$

Coupling-factor denominator (with $\alpha \approx 1/137$):

$$4\pi\alpha = 4\pi/137.$$

Evaluate:

$$4\pi\alpha = 0.0917.$$

Divide:

$$\frac{\lambda_e^2}{4\pi\alpha} = \frac{8.41 \times 10^{-12}}{0.0917}.$$

Evaluate:

$$\frac{\lambda_e^2}{4\pi\alpha} = 9.17 \times 10^{-11}.$$

Multiply by the propagator ratio:

$$R_H^{\max} = 9.17 \times 10^{-11} \times 0.719.$$

$$\boxed{R_H^{\max} \approx 6.6 \times 10^{-11}.$$

Comment. $|R_H|^2 \sim 4 \times 10^{-21}$: the Higgs exchange amplitude is suppressed by $\sim 10^{-10}$ relative to QED, and its contribution to the rate by $\sim 10^{-20}$. Even with QED interference ($\sim 2R_H$) the relative shift is $\sim 10^{-10}$ — far below experimental precision in $ee \rightarrow ee$. Direct Higgs measurement via t -channel exchange is impractical; one instead uses on-shell production (e.g. $H \rightarrow ZZ^*$, $H \rightarrow \gamma\gamma$) where σ peaks at the Higgs mass.